

Computer Vision Assignment

Code :- ITA0502

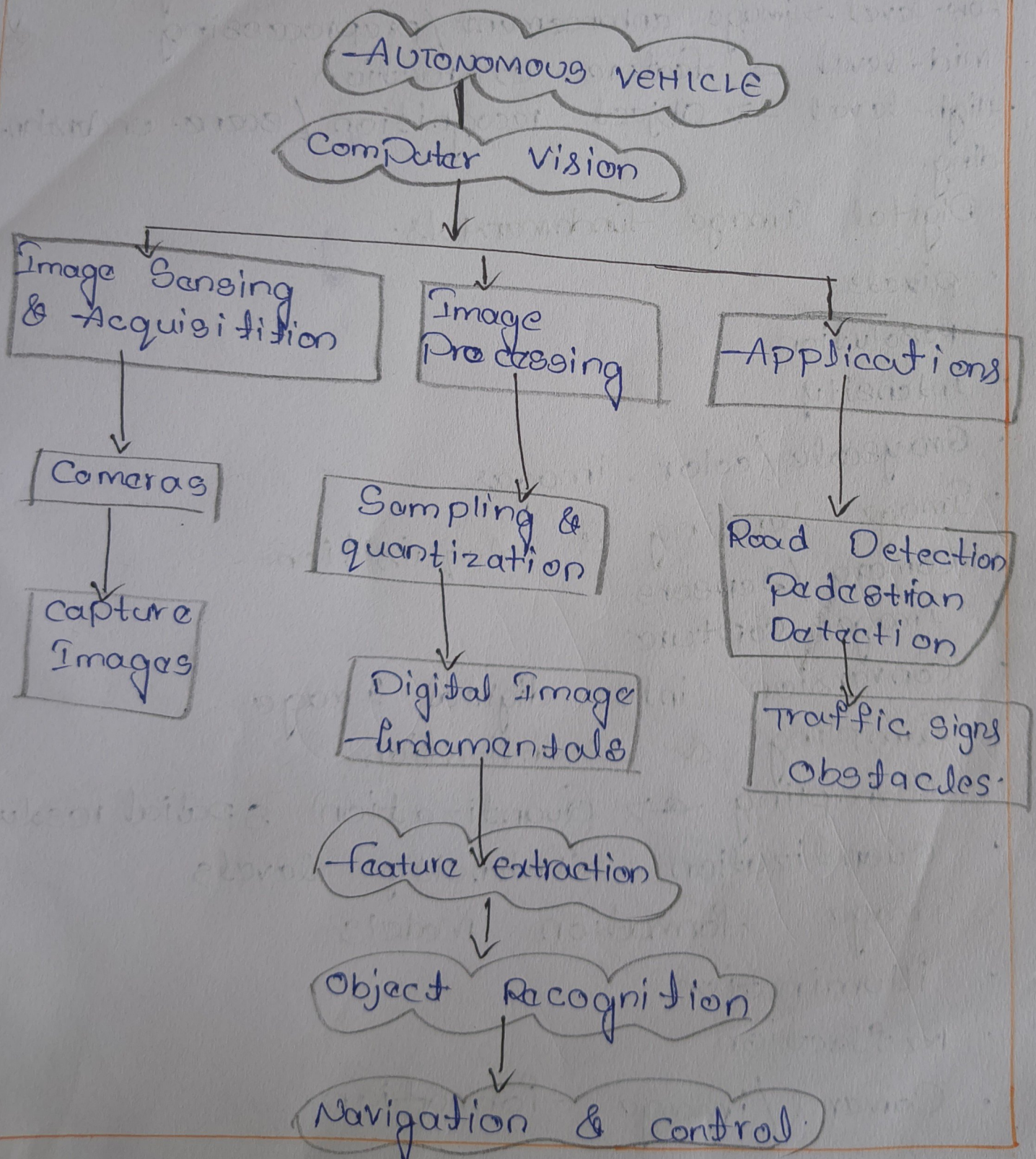
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Course :- Computer Vision.

1. Autonomous vehicle - Concept map

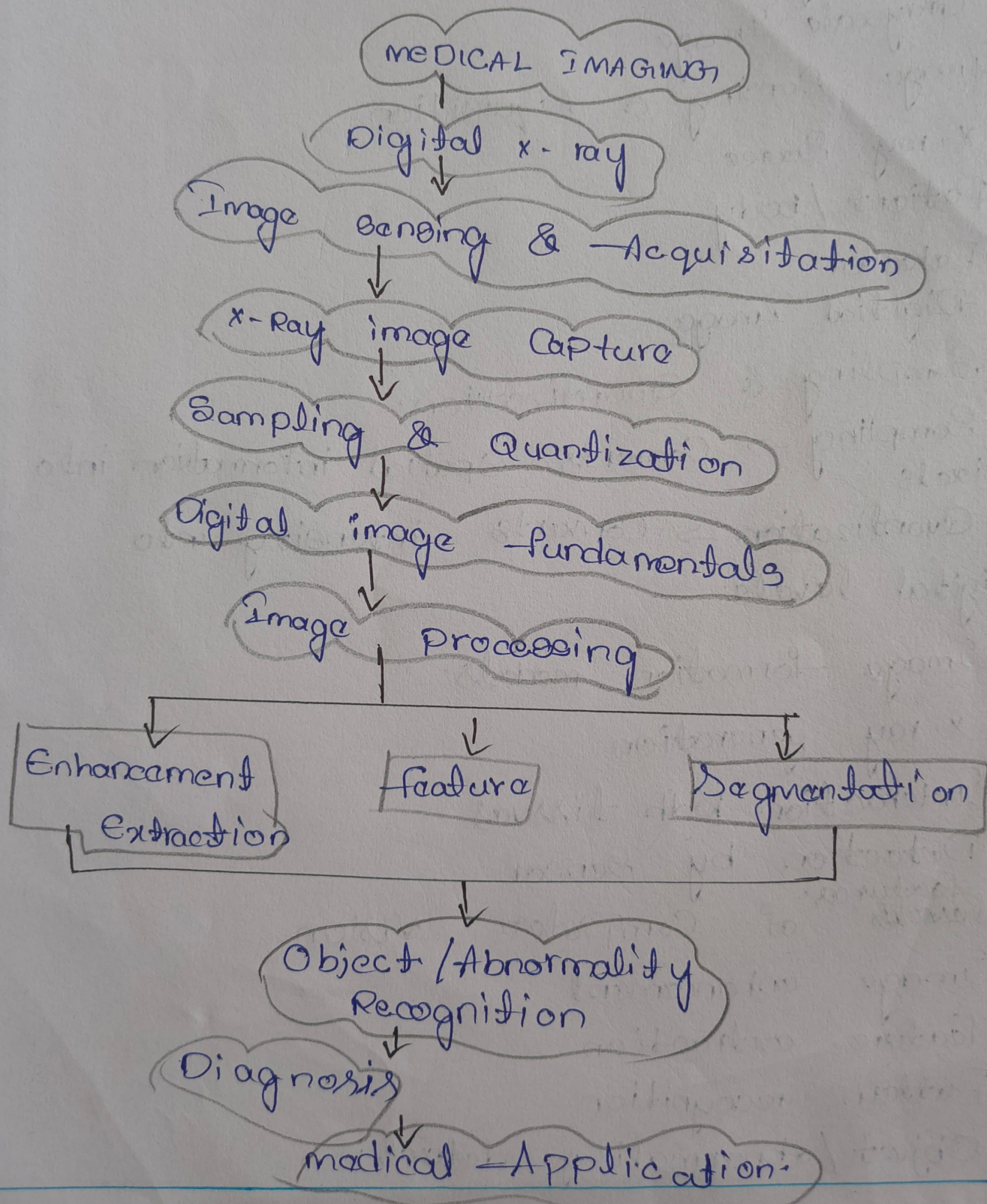
Autonomous Vehicle $\rightarrow$ Computer Vision $\rightarrow$ Image Capture $\rightarrow$ Image Processing $\rightarrow$ Feature Extraction $\rightarrow$ Object Recognition $\rightarrow$ Decision Making $\rightarrow$ Navigation & Control.



- Overview & features of Computer Vision.
- Image understanding
- Object detection
- Pattern recognition
- Scene interpretation.
- Levels of Computer Vision & Image Processing
 - Low-level $\rightarrow$ image enhancement/preprocessing.
 - Mid-level $\rightarrow$ feature extraction
 - High-level $\rightarrow$ Object recognition / scene understanding.
- Digital Image fundamentals.
 - pixels
 - Resolution
 - Intensity
 - Grayscale / color images.
- Image sensing & Acquisition.
 - Camera / sensors
 - Image Capture
- Conversion into digital Image.
 - Sampling & Quantization.
 - Sampling $\rightarrow$ (Quantization) spatial resolution
 - Quantization $\rightarrow$ Intensity levels
- Image formation models.
 - illumination
 - Reflection
 - Camera / image formation.

2. medical Imaging - Concept map

medical Imaging $\rightarrow$ x-Ray Image Acquisition
 $\rightarrow$ Digitization $\rightarrow$ Image processing $\rightarrow$ feature
Extraction $\rightarrow$ Abnormality Detection $\rightarrow$ Diagnosis



Topic to include:-

- Digital Image fundamentals
- Pixel
- Resolution
- Intensity
- Grayscale Image
- Image Sensing & Acquisition.
 - X-ray Source
 - Patient / body
 - Detector
 - Digital Image
- Sampling & Quantization.
 - Sampling $\rightarrow$ Converts Spatial information into Pixels
 - Quantization $\rightarrow$ Converts intensity into digital levels.
- Image formation models.
 - X-ray generation
 - Interaction with tissues
 - Detection by sensor.
- ~~Research~~ ^{features} of Computer Vision.
 - Image enhancement
 - Feature extraction
 - Pattern recognition
 - Object / abnormality detection